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Jannik Laval (Université de Lyon 2, DISP laboratory)

Sylvain Guérin (IMT-Atlantique, DISP laboratory)

Subject: Application for PhD Position – Aggregation of digital twins in the manner of Systems of Systems (PC2)

Dear Jannik Laval and Sylvain Guérin,

I am writing to express my strong interest in the PC2 PhD position "Aggregation of digital twins in the manner of Systems of Systems" at the DISP laboratory. With my background in Artificial Intelligence, Graph Neural Networks, and my hands-on experience building multi-layered digital twin architectures, I am highly motivated to contribute to the Artemis open-source framework and address the fundamental challenges of DT composability and aggregation.

Your objective to construct Digital Twins as modular Systems of Systems (SoS) perfectly aligns with my engineering philosophy. In my recent work architecting an Industrial Gas Turbine Digital Twin, I faced firsthand the challenges of ensuring modularity and interoperability across complex architectures. By combining physical thermodynamic models (e.g., Brayton Cycle) with Edge-TCN anomaly detection, I developed a scalable system that mirrors the operational states of the actual twin (AT). This experience has given me deep insights into the structural and granular requirements necessary to ensure continuous adaptability and high-fidelity representation of the AT throughout its lifecycle. Applying these lessons to define robust composability semantics across heterogeneous models will be my primary focus.

Furthermore, my expertise in deep learning and automated systems positions me well to define the rigorous mathematical and structural rules required for federating independent DT components. I am eager to apply these skills within the DISP laboratory to solve the fundamental metamodeling challenges you have outlined, ensuring functional independence while contributing to the France 2030 EDT initiative. Please find attached my resume and transcripts. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui